

TIME AND WORK – 1

Directions for question 1: Select the correct alternative from the given choices.

A can do a piece of work in 12 days and B can do the same work in 18 days. If A works for three days, in how many days can the remaining work be completed by B?

- ☐ a) $13\frac{1}{2}$
- ☐ b) 9
- ☐ c) $16\frac{1}{2}$

☒ d) 12 (Your answer - ✖)

Solution:

Assume total work = LCM (12, 18) = 36.

If the entire work is 36 units, A's rate is 3 units/day and B's rate is 2 units/day.

In 3 days, A completes 9 units. B would take $\frac{27}{2}$, i.e. 13.5 days to complete the remaining 27 units.

Choice (A)

Directions for question 2: Type in your answer in the input box provided below the question.

There are 60 volumes of a literary work. Suresh can read five volumes in one week and Mohan can read seven volumes in one week. If they start simultaneously and read separately, in how many weeks will they together cover the 60 volumes?

Your Answer: 5 ✔

Directions for question 3: Select the correct alternative from the given choices.

Three workers transfer a tool weighing 120 kg in 12 hours. How many men are required to transfer it in 9 hours?

☒ a) 4 (Your answer - ✓)

☐ b) 5

☐ c) 6

☐ d) 8

Directions for question 4: Type in your answer in the input box provided below the question.

A and B can complete a work in 35 days. A alone can complete the work in 60 days. Find the number of days in which B alone can complete the work.

Your Answer: 16.8 ✗

Solution:

As A and AB take 60 and 35 days, their rates are in the ratio 7 : 12. If the work is taken as 420 units ($\therefore$ LCM (60, 35)), their rates would be 7, 12 respectively, i.e. B's rate would be 5. He would take 84 days to complete 420 units. Ans : (84)

Directions for question 5: Select the correct alternative from the given choices.

A can do a piece of work in 18 days. He worked for 12 days and left. B finished the remaining work in 8 days. In how many days can B alone complete the work?

☒ a) 21 (Your answer - ✗)

☐ b) 22

☐ c) 23

☐ d) None of these

Solution:

As A needs 18 days for the entire work, he completes $\frac{2}{3}^{\text{rd}}$ of the work in 12 days.
To complete the remaining $\frac{1}{3}^{\text{rd}}$, B takes 8 days.
 $\therefore$ B would take 24 days to complete the entire work. Choice (D)

Directions for question 6: Type in your answer in the input box provided below the question.

A is thrice as efficient as B and is therefore able to finish a work in 50 days less than B. Find the number of days taken by them to complete the work, working together.

Your Answer: 10 ✖

Correct Answer: 18.75 ✔

Solution:

A's rate is 3 times that of B. The times that they take is in the ratio 1:3. As A takes 50 days less than B, A takes 25 days and B takes 75 days. If they work together, they would take $\frac{75}{4}$, i.e. 18.75 days. Ans : (18.75)

Note: If A takes x hours to complete a job and B takes y hours to complete the same, they'll together take $\frac{xy}{x+y}$ hours.

Directions for question 7: Select the correct alternative from the given choices.

24 persons can construct a wall in 15 days, but the owner wants it to be completed in 12 days. How many more men should he employ?

☐ a) 4

☒ b) 6 (Your answer - ✔)

☐ c) 8

☐ d) None of these

Directions for question 8: Type in your answer in the input box provided below the question.

A father can do a job twice as fast as his two sons put together. If one son can do it in 6 hours, and the other can do it in 12 hours, in how many hours can the father do the job?

Your Answer: 2 ✓

Directions for question 9: Select the correct alternative from the given choices.

Three men or four women can construct a hut in 31 days. How long will it take 12 men and 15 women to do the same job?

- ☐ a) 8 days
- ☐ b) 12 days
- ☐ c) 4 days

☒ d) None of these (Your answer - ✕)

Solution:

Three men or four women can construct the hut in 31 days. If the work is W,
 $W = 3 (31) \text{ mandays} = 4 (31) \text{ womandays}$.

$\therefore$ one man is equivalent to $\frac{4}{3}$ rd of a woman.

12 men + 15 women are equivalent to 16 women + 15 women.
They would need 4 days to complete 121W.

Choice (C)

Directions for question 10: Type in your answer in the input box provided below the question.

A man, a woman and a boy can do a work in 20 days, 30 days and 60 days respectively. How many boys must assist 2 men and 8 women to do the work in 2 days?

Your Answer: 8 ✓

Directions for questions 11 to 14: Select the correct alternative from the given choices.

A and B can individually complete a work in 8 days and 12 days respectively. If A works at half his usual rate and B at one-third his usual rate, how long will they take to complete the work together?

☐ a) 12 days

☒ b) $11\frac{1}{13}$ days (Your answer - ✓)

☐ c) $13\frac{1}{13}$ days

☐ d) $10\frac{10}{11}$ days

Directions for questions 11 to 14: Select the correct alternative from the given choices.

A cashier in a bank can count 100 notes in one minute. A counting machine can do the same job in 20 seconds. If they work together, how long does it take to count 100 notes?

☐ a) 12 seconds

☐ b) 14 seconds

☒ c) 15 seconds (Your answer - ✓)

☐ d) $16\frac{2}{3}$ second

Directions for questions 11 to 14: Select the correct alternative from the given choices.

A bath tub can be filled by one pipe in 4 minutes and emptied by another in 5 minutes. If the tub is empty and both pipes are opened simultaneously, how long will it take for the tub to be full?

☐ a) 18 minutes

☒ b) 20 minutes (Your answer - ✓)

☐ c) 15 minutes

☐ d) None of these

Directions for questions 11 to 14: Select the correct alternative from the given choices.

Two pipes P and Q can fill a tank in 12 minutes and 16 minutes respectively. If both the pipes are opened simultaneously, then after how many minutes should Q be closed so that the tank is full in 9 minutes?

☐ a) 1

☐ b) 2

☐ c) 3

☒ d) 4 (Your answer - ✓)

Directions for question 15: Type in your answer in the input box provided below the question.

Three men A, B and C can do a work together in 10 days. If A alone can do it in 30 days and B alone in 45 days, how many days would C alone take to complete the work?

Your Answer: 22.5 ✓